
Funmilayo Egunjobi
MSc (Zoo), BSc (Ed) (Bio/Bio Ed), NCE (Bio Ed)

RESEARCH AND TEACHING SUMMARY

I specialize in molecular insect toxicology and vector biology. My research focuses on elucidating the molecular mechanisms of insecticide action and resistance in disease-vector mosquitoes, particularly the interaction of pyrethroids with voltage-gated sodium channels. I integrate molecular, physiological, and computational approaches to identify resistance-associated mutations and advance structure-based insecticide evaluation for sustainable vector control strategies. In addition to maintaining active research, I am an experienced educator and mentor with over 6 years of undergraduate teaching experience, emphasizing inclusive, student-centered learning.

EDUCATION

Doctor of Philosophy (Ph.D.) in Biology Duke University, Durham, NC	2021-present
Master of Science (M.Sc.) in Zoology & Environmental Biology University of Nigeria, Nsukka, Nigeria <i>Summa cum laude</i>	2019
Bachelor of Science (B.Sc.) in Biology & Biology Education Tai Solarin University of Education, Ogun State, Nigeria Valedictorian / <i>Summa cum laude</i>	2016
Nigeria Certificate in Education (NCE) in Biology Education Federal College of Education, Ogun State, Nigeria <i>Summa cum laude</i>	2009

TEACHING / RESEARCH APPOINTMENTS

Graduate Research Assistant (Insect Neurophysiology and Neurotoxicology) Ke Dong Lab, Department of Biology <i>Duke University, Durham, NC, United States</i>	Spring 2026 - present
Graduate Teaching Assistant (Animal Physiology) Department of Biology <i>Duke University, Durham, NC, United States</i>	Spring 2024
Graduate Teaching Assistant (Molecular Biology) Department of Biology <i>Duke University, Durham, NC, United States</i>	Fall 2023
Graduate Teaching Assistant (Microbiology) Department of Biology <i>Duke University, Durham, NC, United States</i>	2022 - 2023
Assistant Lecturer of Biology Department of Biological Sciences <i>Tai Solarin University of Education, Ijebu-Ode, Nigeria</i>	2020 - 2021

Graduate Teaching Trainee Department of Biological Sciences <i>Tai Solarin University of Education, Ijebu-Ode, Nigeria</i>	2018 - 2020
Teaching Assistant Delta State University, Nigeria. <i>National Youth Services Corps, Federal Government of Nigeria</i>	2016 - 2017
Teacher Trainee Our Lady of Apostles Secondary School, Ijebu-Ode, Nigeria	2015
Teacher Trainee Baptist Girls' College, Ogun State, Nigeria	2008 - 2009

FUNDING AND AWARDS

PBT Graduate Student Travel Award Entomological Society of America (Physiology, Biochemistry, and Toxicology Session)	2025
Conference Travel Award The Graduate School, Duke University, Durham, NC, United States	FY 2025 – 2026
Department of Biology One Semester Fellowship Duke University, NC, USA	Fall 2025
Margaret Webster Fellowship Fund Duke University, NC, USA	Spring 2025
Department of Biology Graduate Student Fellowship Fund Duke University, NC, USA	Fall 2024
Jorge Arias Student Competition Award for first-place poster winner Virginia Mosquito Control Association, Virginia, USA	January 2024
Conference Travel Award The Graduate School, Duke University, Durham, NC, United States	FY 2024 – 2025
President's prize for the best poster presentation, Second-place winner Entomological Society of America 2023 conference, Maryland, USA	November 2023
Summer Research Fellowship The Graduate School, Duke University, Durham, NC, United States	Summer 2023
Summer Research Fellowship The Graduate School, Duke University, Durham, NC, United States	Summer 2022
Graduate Student Ph.D. Research Award The Graduate School, Duke University, Durham, NC, United States	2021- 2028
Dr. Sylvia Meek's Scholarship Award for Entomology Malaria Consortium, London, United Kingdom	2017-2019

Automatic Teaching Employment Award Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
Overall Best Graduating Student Award, <i>Summa cum laude</i>, and Valedictorian Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
University Award for the Overall Best Student Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
The Pro-Chancellor's Award for the Overall Best Student Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
The Governor's Award for the Overall Best Student Ogun State Government of Nigeria	2016
University Alumni Prize for the Overall Best Student Alumni Association, Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
College of Science and Information Technology Award Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
Departmental Prize for the Overall Best Student Department of Biology, Tai Solarin University of Education, Ijebu-Ode, Nigeria	2016
Biyi Sangosanya Prize and Award All Nigeria Confederation of Principals of Secondary Schools, Nigeria	2016

PUBLICATIONS

Funmilayo Egunjobi, Felipe Andreazza, Boris S. Zhorov, and Ke Dong. (2024). A Unique Mechanism of Transfluthrin Action Revealed by Mapping Its Binding Sites in the Mosquito Sodium Channel. *Insect Biochemistry and Molecular Biology*, 175, 104214.

Egunjobi, F.B. (2023). Learning how to learn effectively. *American Entomologist*, 69(2), 28–29.

Adewoga, T., Adenusi, A., Okedere, E., Adediran, O., Adebambo, A., Banjo, F., Uthman, T., Adefuye, A., & **Egunjobi, F.B.** (2021). Biodiversity of mosquito larval habitats and its public health implication in communities of Tai Solarin University of Education, Ijagun, Ijebu-Ode, Ogun State. *Afr. J. Med. Med. Sci.* 50, 487-493.

Ojo, F.T. & **Egunjobi, F.B.** (2021). Larvicidal activities of *Ocimum basilicum* and *Mentha piperita* extracts against *Aedes mosquito* larvae. TASUED Journal of Pure and Applied Sciences. A College of Science and Information Technology publication, Tai Solarin University of Education, Ijagun, Ogun State.

Egunjobi, F. B. & Okoye, I. C. (2020). Ovicidal and Larvicidal Activities of Ethanolic Leaf Extracts of Three Botanicals against the Malaria Vector - *Anopheles gambiae*. *International Annals of Science* 9(1): 111-121. <https://doi.org/10.21467/ias.9.1.111-121>

Adewoga, T.O.S., Adediran, O. A., Adenusi, A.A., Adefuye, A. L., Logunleko, O. A., **Egunjobi, F. B.** and

Ajayi, O. T. (2020). Schistosomiasis risk behaviors among primary school pupils and alternative medicine therapy in endemic areas of Ogun State. *Cepress Journal of Science and Innovative Studies*, 1: 126-131

Awofodu, A.D., Saka, V.O, Olanrele, O.J. & **Ogunkomaya, F.B.** (2020). Influence of cultural practice-related misconceptions on achievement of senior secondary biology students. *African Journal of Science, Technology and Mathematics Education*, 5(1), 57-63. A publication of the Department of Science Education, University of Nigeria, Nsukka.

Awofodu, A. D., Azeez, R. O., Mohammed, A. A., Obuh, C. A., Ade-Ogunnowo, F. E., Fabinu, E. O. & **Ogunkomaya, F. B.** (2018). Game strategy on students' learning of genetics and evolution: a means of national development. *TASUED Journal of Pure and Applied Sciences*, 1(1), 25-33. A Publication of the College of Science and Information Technology. Tai Solarin University of Education

CONFERENCES & PROFESSIONAL PRESENTATIONS

Egunjobi, F. (2025). *Knockdown-resistant mutations confer differential sensitivities to structurally distinct pyrethroids*. Oral presentation at the **“Breaking Barriers in Insecticide Resistance: Honoring Prof. Jeff Scott's Career in Insect Toxicology” Symposium**, Entomological Society of America Annual Conference, Portland, Oregon (Nov 9–12).

Egunjobi, F. (2024). *Transfluthrin exhibits a distinct mechanism of action on the mosquito sodium channel*. Invited talk in the session **Identifying & Developing New Tools for the Sustainable Control of Disease Vectors**, American Chemical Society Conference, Fall 2024, Denver, CO (Aug 18-22).

Egunjobi, F. (2024). *Identification of sodium channel mutations conferring pyrethroid resistance in Aedes aegypti mosquitoes*. Poster presentation, Virginia Mosquito Control Association Annual Conference, Hilton Virginia Beach Oceanfront, VA, February 20–22, 2024.

Egunjobi, F. (2023). *Functional characterization of five recently discovered kdr mutations in Aedes aegypti mosquito sodium channels*. Poster presentation, Entomological Society of America Annual Conference, National Harbor, MD, 2023.

MENTORSHIP

Research Mentor, Biology Undergraduate Independent Research Study	Spring 2026
Shankul Lohakare, Duke University, Durham, NC, United States	

Research Mentor, Biology Undergraduate Independent Research Study	Spring 2026
Eubey Kang, Duke University, Durham, NC, United States	

Research Mentor, Undergraduate Semester Research Program	Fall 2025
Wenrui Wu, Duke Kunshan University, Jiangsu, China	

Research Mentor, Biology Undergraduate Independent Research Study	Fall 2023
Ishaan Narsinghani, Duke University, Durham, NC, United States	

Research Mentor, Biological Sciences Undergraduate Research Fellow,
Arriella Ruiz, Duke University, Durham, NC, United States

Summer 2022

PROFESSIONAL SOCIETIES

Entomological Society of America
American Chemical Society
Virginia Mosquito Control Association
Malaria Consortium (UK)